Automated Colour-Based Particle Analysis of Cement Raw Meals Microscopic Images Using Machine Learning

25-26J-250

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology

Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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Declaration

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.
The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date: 2025/08/29

Signature of the co-supervisor: Date: 2025/08/29

Acknowledgement

I would like to express my deepest gratitude to my supervisor, Dr. Dinuka Wijendra, for his invaluable guidance, insightful feedback, and unwavering support throughout the duration of this research. His expertise and encouragement were instrumental in shaping this project.

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Finally, I wish to acknowledge the support of all the lecturers and staff who contributed directly or indirectly to the completion of this work.

Abstract

In cement production, the quality of raw meal—a finely ground mixture of materials such as limestone, iron ore, and laterite—is essential for producing high-performance clinker and cement. Traditional quality control methods rely on manual optical microscopy to classify raw meal particles by colour and texture, which is time-consuming, subjective, and prone to errors, especially when processing large volumes of samples. Although advanced techniques like Scanning Electron Microscopy (SEM) provide high accuracy, they are costly and impractical for routine analysis.

Most existing research focuses on clinker analysis or predicting chemical composition, leaving a gap in automated, colour-based particle analysis of raw meal. This study proposes an innovative system that uses machine learning (ML) and computer vision to automate the segmentation, classification, and quantification of raw meal particles from optical microscopy images. By leveraging Convolutional Neural Networks (CNNs) combined with image preprocessing techniques such as colour space conversion (from RGB to LAB or HSV), the system aims to accurately identify mineral particles based on their colour and structural features.

A hybrid CNN model with fully connected layers will be employed to robustly extract features and classify particles, eliminating the need for manual inspection. Designed to be scalable, reliable, and operator-independent, this solution facilitates real-time quality control, reducing compositional fluctuations and enhancing kiln efficiency. Preliminary results indicate that this approach can significantly improve the consistency and efficiency of raw meal characterization, offering a transformative tool for optimizing cement production processes.

Table of content

Declaration i
Acknowledgement ii
Abstract iii
Table of content iv
List of figures vi
List of tables vi
1. Introduction 1
1.1 Background & Literature survey 2
1.1.1 Microscopy in Cement Analysis 2
1.1.2 Homogenization & Composition of Raw Meal 8
1.1.3 Machine Learning in Colour-Based Particle Analysis of Raw Meal (Cement) 9
1.2 Research Gap 11
1.3 Research Problem 12
2. Objectives 13
2.1 Main Objectives 13
2.2 Specific Objectives 13
3. METHODOLOGY 14
3.1 System Architecture Diagram 14
.... 15
3.2 Understanding the main components of research area 16
3.2.1 Data Collection 16
3.2.2 Image Preprocessing 16
3.2.3 Model Development 16
3.2.4 Performance Evaluation 17
3.2.5 Integration into INSEE's Quality Control 17
3.2.6 Pilot Testing and Refinement 18
4. Project Requirements 18
4.1 Functional Requirements 18
4.2 Non-Functional Requirements 19
4.3 System Requirements 20

4.4 Work Breakdown Structure 22
5.Grant chart 23
6. Budget and Budget Justification 24
7. Commercialization Strategy 25
8. Reference list 27
9. APPENDICIES 28

List of figures

Figure 1survey result clinker particle identification and quality checks......................3
Figure 2 : survey result cement familiar..3
Figure 3 survey result colour-based particle identification...4
Figure 4 survey result improve safety...4
Figure 5 survey result strengths prediction...5
Figure 6 survey result reduce workload..5
Figure 7survey result features like..6
Figure 8 survey result time spend on manual microscopy...6
Figure 9 survey result AI based recommendation...7
Figure 10 survey result help of AI driven tool at work..7
Figure 11 system overview diagram..14
Figure 12 system diagram for raw meal particle analysis..15
Figure 13 work breakdown structure...22
Figure 14 grant chart..23

List of tables

Table 1 research gap..11
Table 2 budget table...24

1. Introduction

The production of high-performance cement depends heavily on consistently maintaining the quality of raw materials. In the cement industry, materials like limestone, iron ore, alumina, and laterite are mixed and ground into a fine, uniform blend called raw meal. This raw meal is what gets fed into the kiln to form clinker. The physical and chemical makeup of this raw meal directly impacts the quality of the clinker and, ultimately, the final cement product.

Traditionally, quality control involves microscopic examination, where operators visually inspect and classify mineral particles based on things like color and texture. For example, particles rich in iron often appear dark red, while quartz particles tend to be lighter or whitish. Although this manual method is common, it is time-consuming, subjective, and prone to errors. especially when there are large numbers of samples to evaluate.

This study proposes an automated solution that uses Machine Learning (ML) and Computer Vision (CV) to analyze color-based microscopic images of raw meal samples. By using Convolutional Neural Networks (CNNs) a powerful deep learning model known for image classification and segmentation. this approach can automatically learn important color and structural features directly from high-resolution images, eliminating the need for manual feature selection.

To improve color detection accuracy, image preprocessing methods such as converting color spaces (from RGB to LAB or HSV), applying histogram equalization, and segmenting images will be used. Clustering techniques like K-Means or Gaussian Mixture Models might also be applied to initially segment particles based on color similarities.

While CNNs excel at image recognition, Recurrent Neural Networks (RNNs), which are better suited for sequential data like time series or text, are not ideal for analyzing static microscope images. Instead, a hybrid approach combining CNNs for feature extraction with fully connected layers for classification can deliver highly accurate identification of mineral types.

Overall, this automated system aims to provide a reliable, consistent, and scalable way to characterize raw meal samples, improving quality control in cement production and greatly reducing the reliance on manual inspection.

1.1 Background & Literature survey

1.1.1 Microscopy in Cement Analysis

Machine learning (ML) and computer vision have increasingly found practical applications throughout various stages of cement production. In particular, there has been growing focus on addressing challenges related to raw meal the finely ground mixture of raw materials fed into the kiln. Recent research efforts have successfully applied ML techniques for tasks such as predicting the chemical composition of raw meal, monitoring its fineness (particle size distribution), and characterizing individual particles. These studies not only demonstrate the potential of these technologies but also offer valuable methods and algorithms that can be adapted for tasks like colour-based particle segmentation and counting using images obtained from optical microscopes.

Microscopy has long been a cornerstone in understanding the microstructure of cement materials. Among different microscopy techniques. Techniques like Scanning Electron Microscopy (SEM), especially when combined with X-ray microanalysis, have proven extremely useful for precisely measuring the composition of clinker minerals such as alite, belite, ferrites, and gypsum. This method allows for direct measurement of surface areas and phase proportions in clinker, often providing more detailed and accurate information than traditional chemical calculations like Bogue's equations, as it captures the actual structural complexity of the material [1].

Although SEM provides high accuracy, optical microscopy using reflected or transmitted light is a more accessible and cost-effective tool. It's particularly helpful for quickly identifying mineral phases in raw meal or clinker samples by revealing important features like colour differences, particle size, and shapes. For example, iron-rich particles typically have reddish hues, while quartz shows up as white or translucent. Optical microscopy also helps researchers understand how certain components, like quartz, affect reactions in the kiln such as influencing belite formation which in turn can impact early strength development and the efficiency of the kiln process [2].

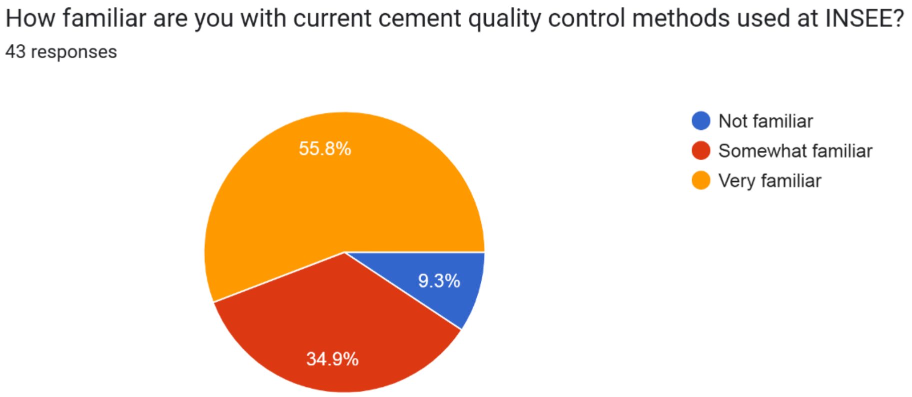

Figure 2 : survey result cement familiar

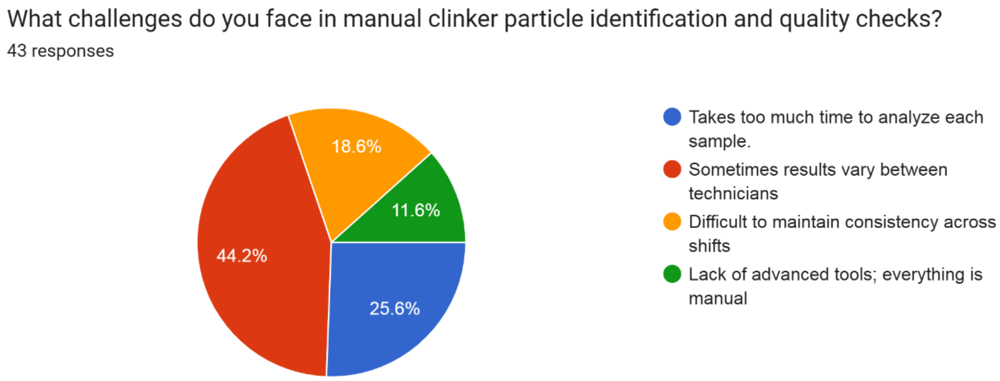

Figure 1survey result clinker particle identification and quality checks

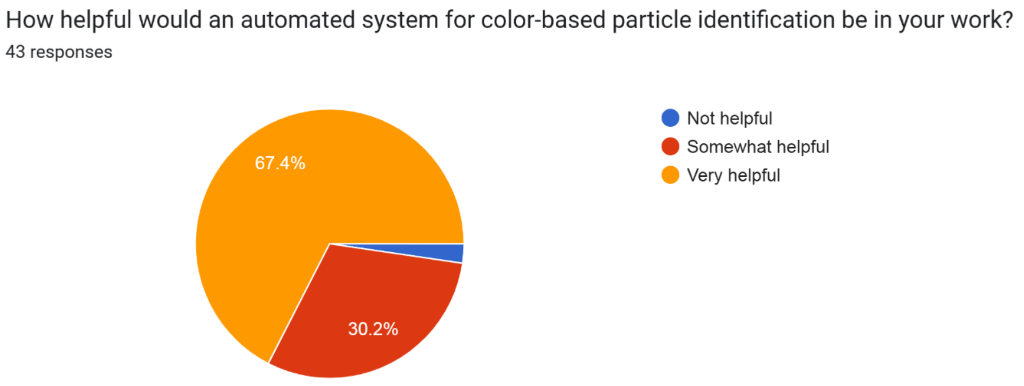

Figure 3 survey result colour-based particle identification

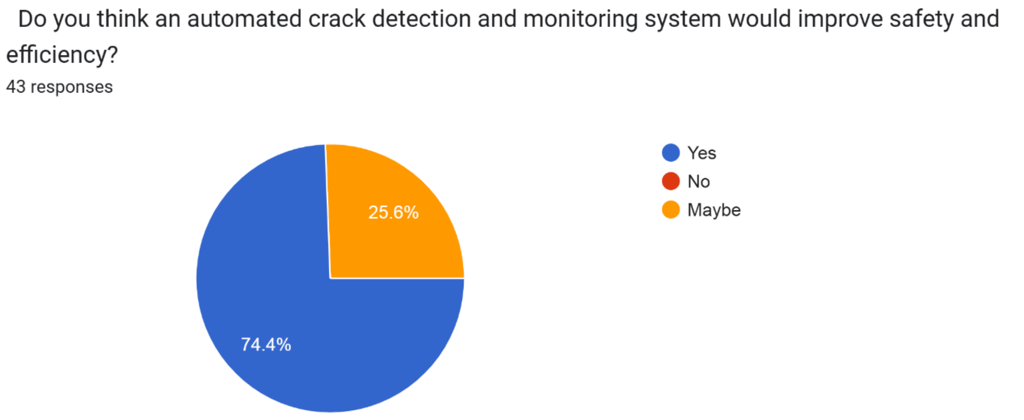

Figure 4 survey result improve safety

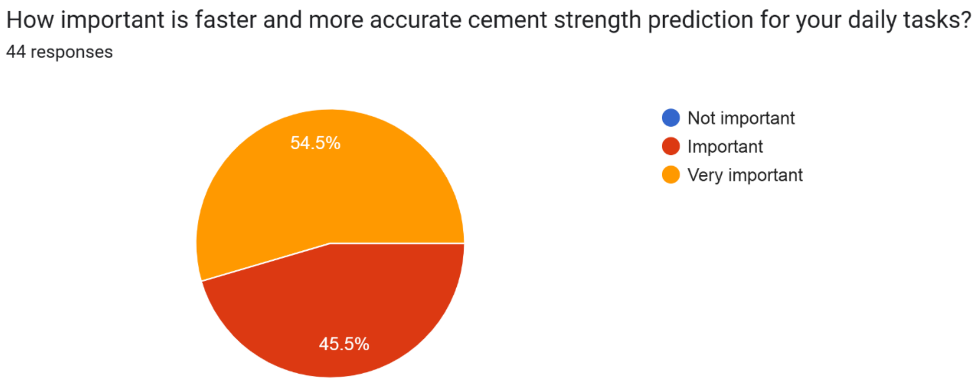

Figure 5 survey result strengths prediction

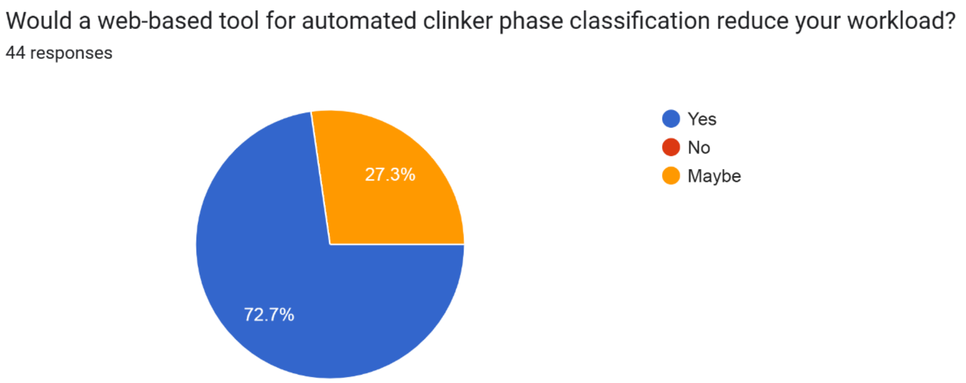

Figure 6 survey result reduce workload

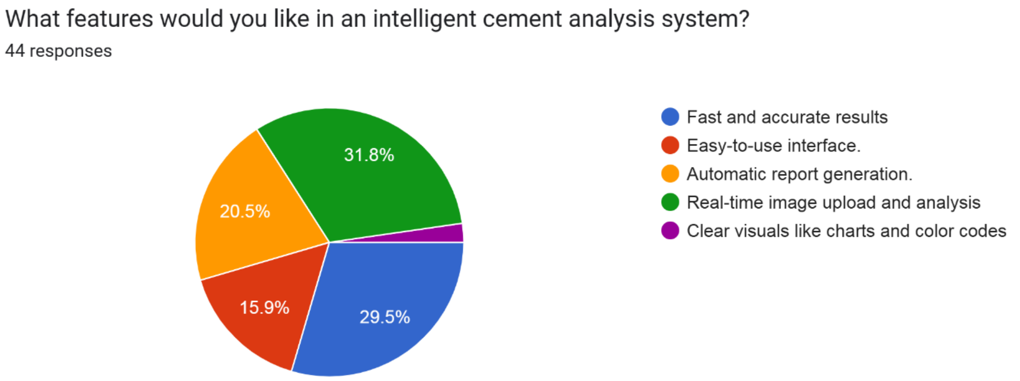

Figure 7survey result features like

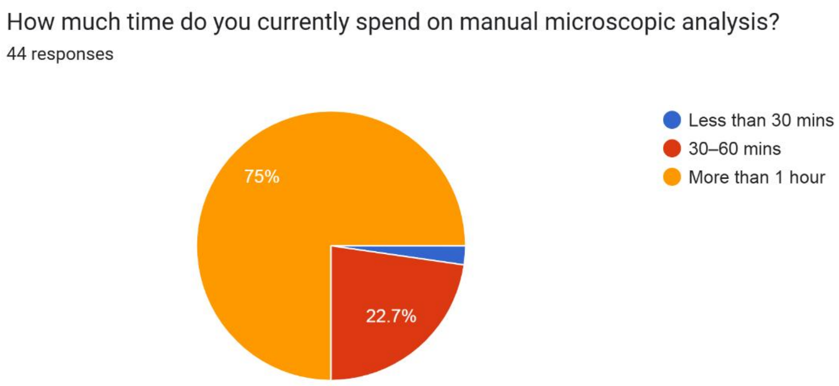

Figure 8 survey result time spend on manual microscopy

What level of trust would you place in AI-based recommendations for cement quality?
45 responses

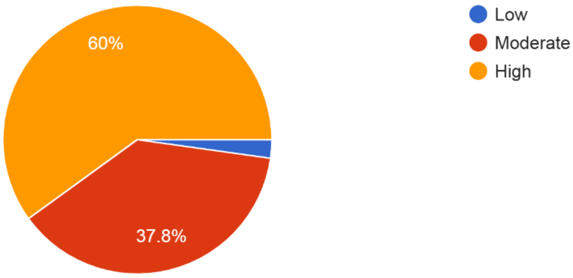

Figure 9 survey result AI based recommendation

Would training or workshops help you adopt AI-driven tools in your work?
45 responses

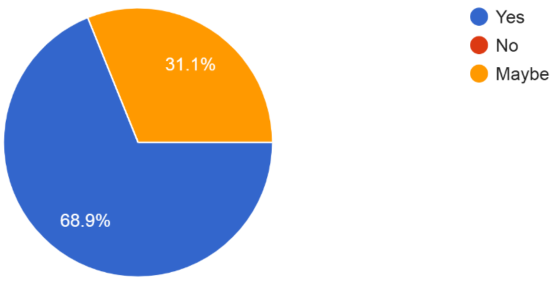

Figure 10 survey result help of AI driven tool at work

1.1.2 Homogenization & Composition of Raw Meal

Ensuring consistency in the chemical composition and thorough mixing (homogeneity) of raw meal is vital for achieving stable kiln operation and producing high-quality cement. Even small fluctuations in the raw meal's composition can cause serious issues in the kiln, such as coating build-up, ring formations, or reduced clinker quality, ultimately impacting product performance and plant efficiency.

To address this, numerous studies have explored various methods for raw meal homogenization. Simulated models of storage and milling equipment measured using techniques like electrical capacitance have shown that pneumatic homogenization silos are far superior to traditional mechanical mixing systems. These silos use air flow combined with gravity to continuously mix multiple layers of raw meal inside the storage silo, promoting effective blending. This multi-layer mixing significantly reduces chemical fluctuations, ensuring a more uniform feed to the kiln and smoother overall operation [3].

In addition to physical homogenization, rapid and accurate chemical characterization of raw meal is essential. Technologies such as Near-Infrared (NIR) spectroscopy, combined with statistical models like Partial Least Squares (PLS) regression, have emerged as powerful tools. These techniques quickly analyze spectra of raw meal samples to predict the concentration of key oxide components (CaO, SiO_2, Al_2O_3, Fe_2O_3) with high accuracy. The prediction quality often equals or surpasses traditional, slower methods like X-ray Fluorescence (XRF) or wet chemical analysis. The speed and reliability of NIR-based methods make them especially suitable for real-time quality control on the production line, enabling timely adjustments to raw meal composition [4].

1.1.3 Machine Learning in Colour-Based Particle Analysis of Raw Meal (Cement)

Recent advances in machine learning (ML) and computer vision have made automated image analysis a powerful and practical tool for studying materials. While much of the existing research has focused on clinker and hardened cement products, there is considerable opportunity to apply ML techniques to raw meal particle analysis, especially for colour-based segmentation, which remains relatively unexplored.

The colour of raw meal particles corresponds closely to their mineral makeup. For example, particles rich in iron tend to appear dark red, while quartz particles are usually white or translucent. Being able to accurately identify and count these colour groups is crucial for monitoring the uniformity of the raw meal and predicting how it will behave during the clinker formation process. However, manually counting particles under an optical microscope is tedious, slow, and can vary significantly depending on the person doing the counting.

Machine learning offers a promising way to automate this task. Convolutional Neural Networks (CNNs), which are well-known for their effectiveness in image recognition and segmentation, can be trained to classify particles based on their colour and texture. Advanced models like Mask R-CNN or U-Net have been successfully used in other materials science applications for segmenting complex images with overlapping particles and intricate boundaries, achieving very high accuracy.

The approach typically starts with compiling a dataset of microscope images of raw meal, where particles are manually labeled by their color categories such as dark red, light red, pink, and white. These labeled images are used to train a CNN-based segmentation model. Once trained, the model can:

- Detect individual particles in microscope images,
- Classify each particle into one of the predefined colour groups,
- Automatically count particles to provide quantitative data on the distribution of colours.

To make the model more robust against variations in imaging conditions, data augmentation techniques like rotation, contrast adjustment, and scaling are used during training. The model's performance is measured using metrics such as

Intersection-over-Union (IOU), precision, recall, and Mean Average Precision (MAP) by comparing its segmentation results to manual counts.

Integrating machine learning into colour-based particle analysis has the potential to revolutionize quality control of raw meal. It can deliver fast, repeatable, and operator-independent results, drastically reducing human error and labor. Beyond counting, this approach also enables more detailed statistical analysis of particle size, colour ratios, and structural patterns information that can help optimize raw meal homogenization and overall cement quality.

1.2 Research Gap

Table 1 research gap

Features	Research paper 1	Research paper 2	Research paper 3	Research paper 4	Proposed System
Raw material particle analysis	No	Yes	Yes	Yes	Yes
Microscopy for color-based Detection	No	No	No	No	Yes
Automated using Machine Learning Approach	No	No	No	No	Yes
Colour-Based Particle Segmentation and Classification	No	No	No	No	Yes

1.3 Research Problem

In cement production, the quality of raw meal a finely ground mix of raw materials like limestone, iron ore, alumina, and laterite is crucial for making high-performance cement. The chemical and physical properties of this raw meal, especially its mineral composition and uniformity, have a direct impact on the quality of clinker formed in the kiln, and ultimately on the final cement product.

Traditionally, quality control involves manually examining raw meal particles under a microscope, classifying them by visual features such as colour and texture iron-rich particles appear dark red, quartz particles are typically white or translucent. However, this manual approach is laborious, slow, and prone to human error, which becomes especially problematic when handling large numbers of samples.

Modern microscopy methods like Scanning Electron Microscopy (SEM) combined with X-ray microanalysis offer very precise clinker analysis. But these techniques are expensive and not practical for routine quality checks of raw meal. Optical microscopy is more affordable but still depends heavily on human interpretation, limiting how quickly and consistently large volumes of samples can be evaluated.

While there has been growing interest in applying machine learning (ML) and computer vision in cement production, most research has focused on clinker analysis, chemical composition prediction, or equipment performance. Few studies have addressed automating the segmentation and classification of raw meal particles based on colour in optical microscope images.

This lack of an automated, reliable, and scalable system to analyze raw meal particles by their colour and structure creates a significant bottleneck in quality control. Such inefficiencies lead to variations in raw meal composition, which can disrupt kiln performance, affect clinker quality, and increase production costs.

Therefore, there is a clear need for an automated solution that leverages machine learning techniques—particularly Convolutional Neural Networks (CNNs) along with image preprocessing methods to accurately segment, classify, and quantify raw meal particles based on their colour and structural features. Implementing this approach would substantially improve the efficiency, repeatability, and accuracy of quality control in cement production.

2. Objectives

2.1 Main Objectives

The main aim of this research is to create an automated system using machine learning to analyse colour-based microscopic images of cement raw meal. This system is intended to improve quality control in cement production by providing a solution that is reliable, scalable, and independent of the operator. By reducing the need for manual microscopy, it aims to minimize errors and enhance the consistency and efficiency of clinker production.

2.2 Specific Objectives

- Develop a Machine Learning Model: Build and implement a Convolutional Neural Network (CNN) model that can automatically segment and classify raw meal particles in optical microscope images based on their colour and structural characteristics.
- Apply Image Preprocessing Techniques: Use methods like colour space conversion (from RGB to LAB or HSV), histogram equalization, and clustering algorithms (such as K-Means or Gaussian Mixture Models) to improve the accuracy of detecting and segmenting particles.
- Quantify Particle Distribution: Enable the system to automatically count particles and analyse their distribution by colour groups (for example, dark red for iron-rich particles and white for quartz) to generate useful quantitative data for quality control.
- Integrate for Real-Time Application: Create a scalable framework that can be integrated into cement production processes, allowing for real-time quality control and helping to reduce fluctuations in the raw meal composition fed into the kiln.

3. METHODOLOGY

This study presents a machine learning and computer vision approach to automate the analysis of cement raw meal particles by using microscopic images gathered from INSEE, a leading cement manufacturer. The methodology covers the entire process from data collection and image preprocessing to building and testing the model, and finally integrating it into INSEE's quality control system. The goal is to achieve accurate, efficient, and scalable particle segmentation, classification, and quantification.

3.1 System Architecture Diagram

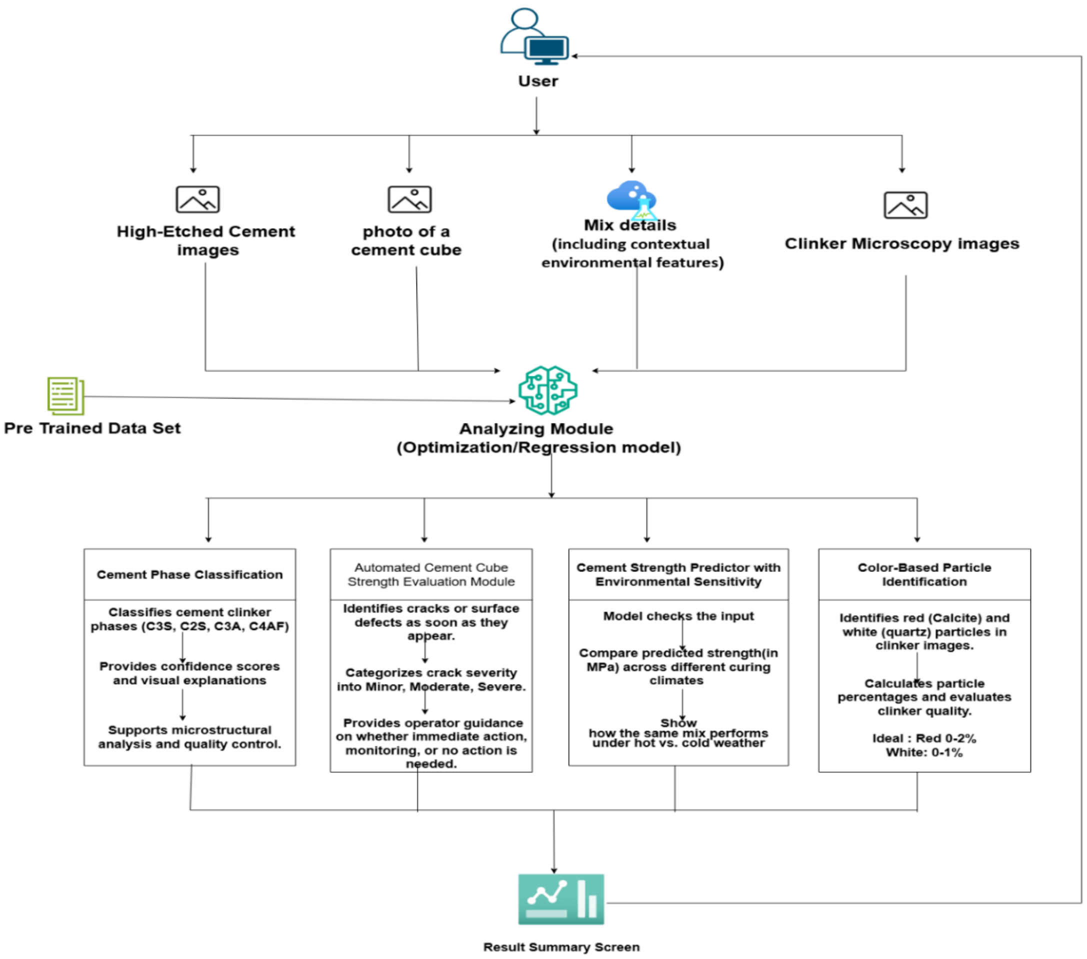

Figure 11 system overview diagram

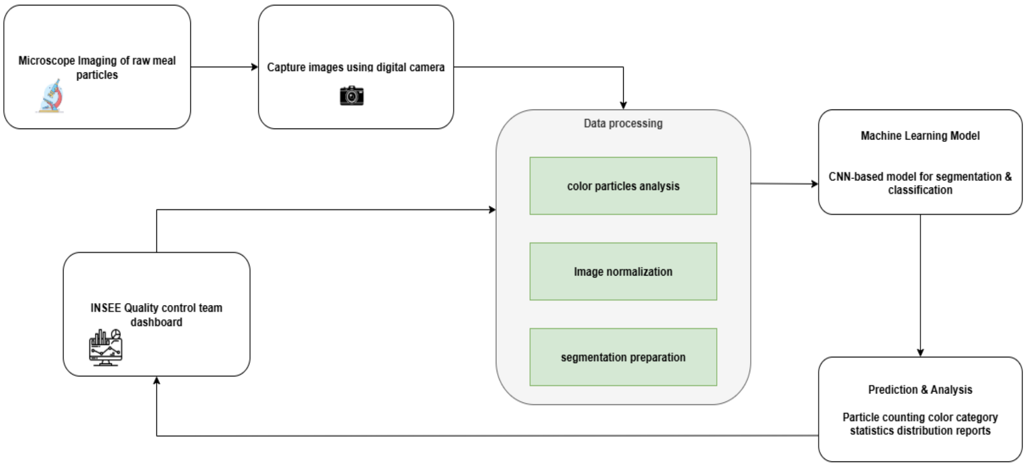

Figure 12 system diagram for raw meal particle analysis

3.2 Understanding the main components of research area

3.2.1 Data Collection

The microscopic images of cement raw meal will be collected at INSEE's production sites using optical microscopes equipped with high-resolution cameras. These images will capture samples containing various minerals like limestone, iron ore, alumina, and laterite, which display distinct colours for example, iron-rich particles appear dark red, while quartz particles are white or translucent.

3.2.2 Image Preprocessing

Images will first be converted from the standard RGB colour space to LAB or HSV colour spaces, which better separate colour from brightness, enhancing the accuracy of colour-based particle detection.

Histogram equalization will be applied to improve contrast, making particle edges and colour differences clearer under varying microscope conditions.

Unsupervised clustering methods such as K-Means or Gaussian Mixture Models (GMM) will be used initially to group particles by similar colours as a preliminary segmentation step.

Advanced segmentation techniques like watershed or thresholding algorithms will then isolate individual particles, helping to address challenges from overlapping particles and complex shapes found in INSEE's samples.

3.2.3 Model Development

A Convolutional Neural Network (CNN) architecture, such as Mask R-CNN or U-Net, will be developed to perform both segmentation and classification of the particles. The CNN will learn to recognize colour and structural features directly from the pre-processed images, removing the need for manual feature selection.

The CNN will be combined with fully connected layers that classify each segmented particle into predefined colour categories (e.g., iron-rich, quartz), ensuring precise and consistent classification.

Training will be done using supervised learning on the labeled dataset, with hyperparameters like learning rate and batch size optimized. Loss functions tailored to segmentation and classification will guide the training process.

Computational resources such as GPU-enabled systems and platforms like TensorFlow or PyTorch will be employed, utilizing INSEE's own infrastructure or cloud services for scalability.

3.2.4 Performance Evaluation

The model's accuracy will be measured using metrics such as Intersection-over-Union (IoU) to assess segmentation and precision, recall, and Mean Average Precision (MAP) for classification performance. These results will be compared to the manual annotations to benchmark accuracy.

A portion of the data (for example, 20%) will be reserved as a validation set to test how well the model generalizes to new samples. Cross-validation methods may also be employed to ensure robustness.

The automated system's output will be compared to INSEE's current manual microscopic analysis to demonstrate improvements in speed, accuracy, and consistency.

3.2.5 Integration into INSEE's Quality Control

The trained model will be incorporated into INSEE's production workflow as a software module that processes microscope images in real-time. It will work alongside INSEE's existing microscopy hardware to analyze raw meal samples during manufacturing.

The system will generate detailed reports showing particle distribution by color and highlight any deviations from expected raw meal composition, helping the production team make quick adjustments.

It will be optimized to handle high sample volumes with minimal computational delay, ensuring smooth operation on the production line.

A user-friendly interface will be developed to display segmented images, classification outcomes, and summary statistics, supporting easy interpretation by quality control staff.

3.2.6 Pilot Testing and Refinement

A pilot study will be carried out at INSEE's facility to test the system's performance in real production conditions. Samples from various batches will be analysed to evaluate reliability and accuracy.

Feedback from INSEE's quality control team will be collected to identify practical challenges, such as dealing with specific particle types or variations caused by source changes.

Based on this feedback and pilot results, the model will be fine-tuned by improving preprocessing methods, retraining with additional data, and updating the CNN architecture to enhance overall performance.

4. Project Requirements

4.1 Functional Requirements

- **Image Acquisition:** It must connect with INSEE's optical microscopes to capture high-quality images of raw meal samples. The system should support different magnifications like 100x and 200x, and accept common image formats such as JPEG and PNG.
- **Image Preprocessing:** The system should enhance the images by:
    - Converting colours from the RGB format to LAB or HSV to better separate colours for analysis.
    - Applying histogram equalization to balance contrast and highlight particle edges.

    - Using clustering methods like K-Means or Gaussian Mixture Models to group particles by colour initially.
    - Employing advanced techniques like watershed or thresholding to clearly separate individual particles, even if they overlap or have complex shapes.

- **Particle Segmentation and Classification:** Using a Convolutional Neural Network (CNN) model (such as Mask R-CNN or U-Net), the system should:
    - Identify and separate each particle in the microscope images.
    - Classify each particle into colour categories predefined by INSEE, for example, dark red for iron-rich particles and white for quartz.
- **Quantitative Analysis:** It should count particles in each colour group and provide statistical summaries like the percentage of iron-rich versus quartz particles.
- **Report Generation:** The system must automatically create reports containing segmented images, classification outcomes, and particle distribution data. These reports should be exportable in formats such as PDF or CSV for review by INSEE's quality control team.
- **Real-Time Integration:** The system should analyse images quickly enough to provide results during the production process ideally within seconds supporting timely decision-making.

- **User Interface:** It should include an easy-to-use interface that allows INSEE's quality control staff to view segmented images, classification outcomes, and summary reports. Users should be able to filter results by sample batch or particle type.

4.2 Non-Functional Requirements

- **Accuracy:** The system should achieve at least 90% accuracy in both particle segmentation and classification. Accuracy will be measured using standard metrics such as Intersection-over-Union (IoU), precision, recall, and Mean Average Precision (MAP), compared against manually labeled data.

- **Performance:** It should be able to process a single high-resolution image (for example, 1920x1080 pixels) within 2 seconds to support real-time quality control on the production line.
- **Scalability:** The system must handle a high volume of images, processing at least 100 images per hour to keep up with INSEE's production requirements.
- **Reliability:** The system should be operational 99% of the time during production hours, ensuring consistent performance.
- **Usability:** The interface should be intuitive enough that quality control personnel can learn to use it effectively with less than 2 hours of training.
- **Robustness:** It must maintain performance even when imaging conditions vary (like changes in lighting or microscope settings) or when raw meal composition changes. This robustness will come from data augmentation and careful preprocessing.
- **Security:** The system must securely store all image data and analysis results, complying with INSEE's data privacy and security policies.

4.3 System Requirements

The backend of the system will be developed using Python, chosen for its versatility in handling image preprocessing and implementing machine learning models.

For data processing and machine learning, several popular libraries will be used:

- Pandas will manage and manipulate data, especially for analysing particle distribution results.
- NumPy will handle numerical calculations and array operations needed during image preprocessing.
- Scikit-learn will be used for unsupervised clustering methods like K-Means and Gaussian Mixture Models, as well as for evaluating model performance using metrics such as precision and recall.

- XGBoost may be employed as an additional technique to boost classification accuracy if necessary.
- TensorFlow will serve as the primary framework to develop and train convolutional neural network models (such as Mask R-CNN or U-Net) for accurately segmenting and classifying particles.
- OpenCV will support image processing tasks including converting colour spaces (e.g., RGB to LAB or HSV), enhancing contrast through histogram equalization, and performing segmentation techniques like watershed and thresholding.

For the web application backend, frameworks such as Flask or Django will be used to build a lightweight, scalable server that handles API calls, processes images, and sends analysis results to the user interface.

The frontend dashboard, designed to be interactive and easy to use, will be built using modern JavaScript frameworks like React.js or Angular. This will allow INSEE's quality control team to view segmented images, classification outcomes, and statistical reports on particle distribution efficiently.

To manage data storage, databases like PostgreSQL or MySQL will be used. These will safely store raw meal image details, labeled datasets, model outputs, and quantitative analysis results, ensuring quick and reliable access to data when needed.

The entire application will be containerized using Docker, which ensures the system runs consistently from development through testing and deployment, whether on INSEE's local servers or cloud platforms.

4.4 Work Breakdown Structure

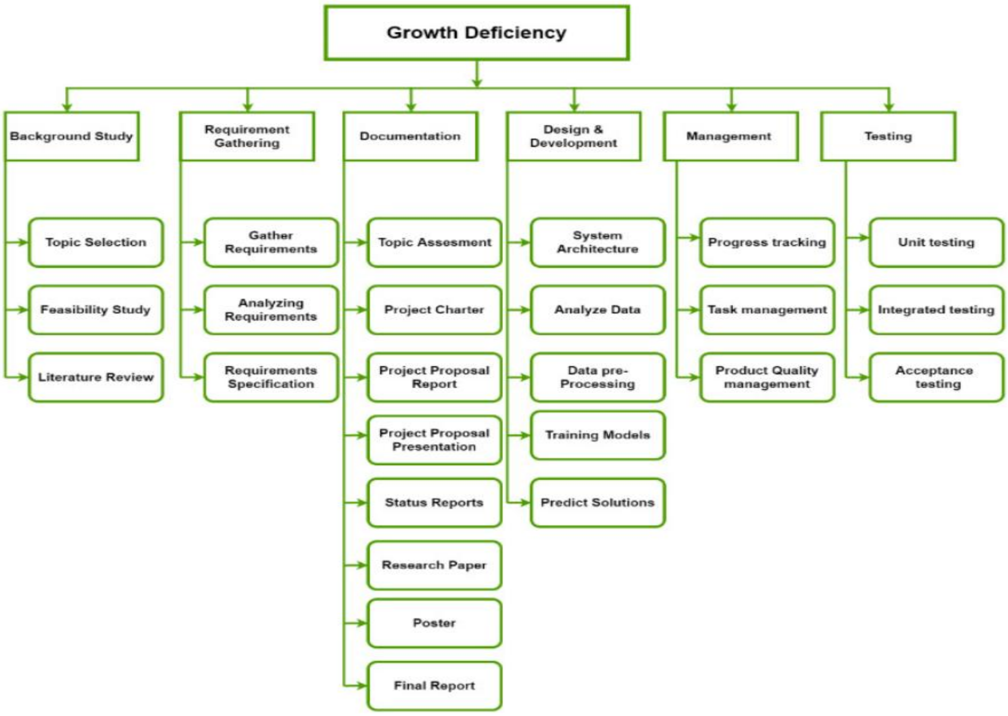

Figure 13 work breakdown structure

5.Grant chart

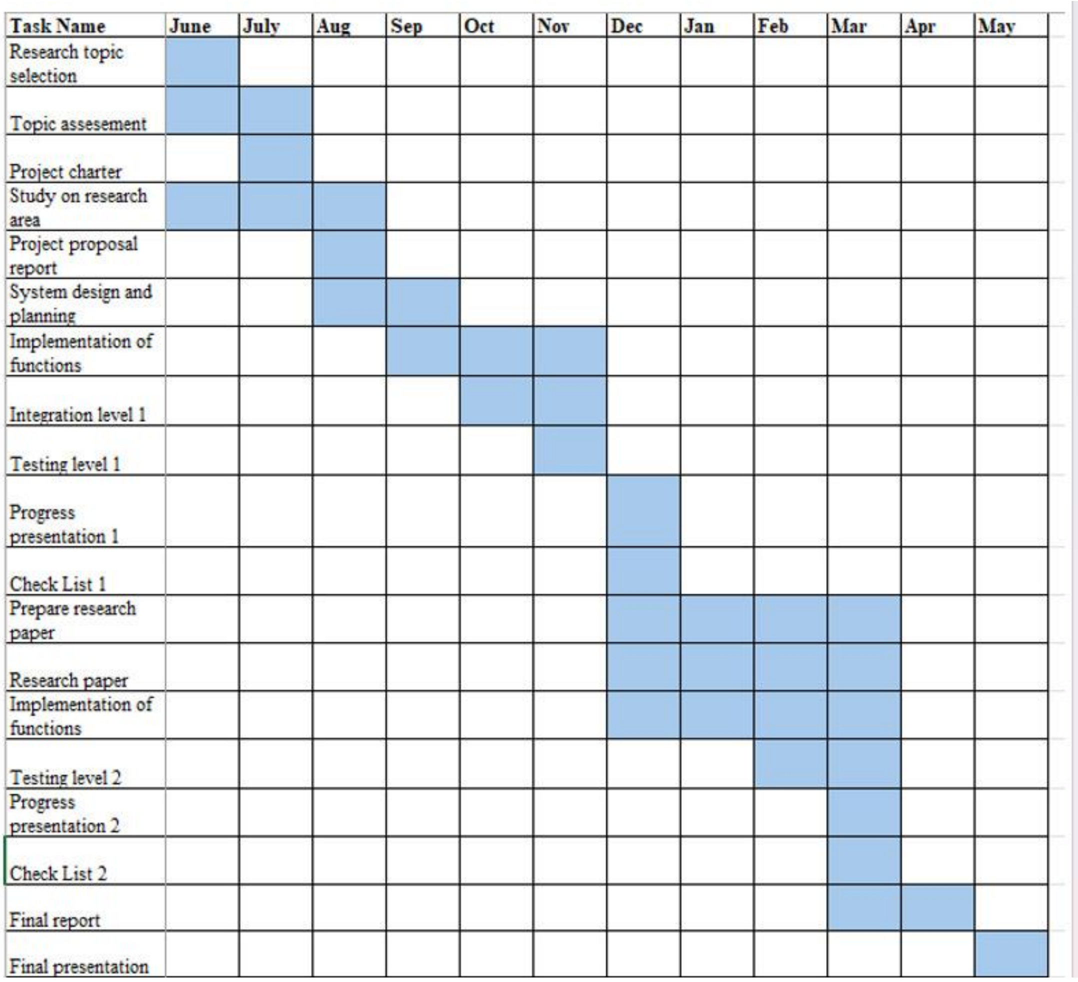

Figure 14 grant chart

6. Budget and Budget Justification

The table below outlines the estimated monthly costs for developing, deploying, and maintaining the Automated Colour-Based Raw Meal Particle Analysis System for INSEE Cement.

Table 2 budget table

Requirement	Cost (Rs.)
System Development and Deployment	15,000 per month
Microscopy Data Acquisition Support	5,000 per 3 months
Machine Learning Model Maintenance	5,000 per 3 months
Total Average	18,333 per month

Budget Justification

- System Development and Deployment:
  This budget covers all expenses related to creating the system using technologies like Python, TensorFlow, Flask or Django for the backend, and React.js or Angular for the frontend. It includes hosting the web interface on INSEE's own servers or cloud services such as AWS, maintaining Docker containers for smooth deployment, and providing IT support to integrate the system with INSEE's existing microscopy hardware. Ongoing updates and improvements will also be handled under this cost to ensure the system operates in real time.
- Microscopy Data Acquisition Support:
  Funds will be allocated for regular calibration and upkeep of the optical microscopes used in capturing images, ensuring consistent high image quality. This also covers tools and databases required for annotating and storing the microscopy data, using systems like PostgreSQL or MySQL.
- Machine Learning Model Maintenance:
  This portion supports periodic retraining of the convolutional neural network models (such as Mask R-CNN and U-Net) to keep them accurate as raw meal compositions vary over time. The maintenance includes using performance feedback from INSEE's quality control team and optimizing the models with evaluation metrics like Intersection-over-Union (IoU) and Mean Average Precision (MAP).

7. Commercialization Strategy

The plan to bring the Automated Colour-Based Raw Meal Particle Analysis System to market for INSEE Cement and other cement producers in Sri Lanka focuses on encouraging adoption, ensuring scalability, and delivering lasting value through improved quality control. The strategy identifies key users, offers different versions of the product to suit various needs, and lays out a clear approach to enter and grow in the cement industry.

Main Target Groups

- Cement Manufacturers (such as INSEE Cement): These are the primary users who will benefit from automated raw meal particle analysis, helping them maintain consistent product quality, optimize kiln performance, and cut production costs.
- Quality Control and Laboratory Staff: The people who will use the system to upload microscope images, get real-time particle classification and distribution data, and make adjustments to raw meal blending accordingly.
- Research & Development Teams: Teams that will use the system to study new raw material compositions, enhance formulation processes, and improve overall efficiency.

Community Version

Designed for small and medium-sized cement plants, this version includes essential tools such as:

- Uploading microscope images for automated detection and classification of raw meal particles by colour.
- Generating straightforward reports showing particle counts and colour ratios to support quality monitoring.
- Providing alerts to notify users if any unusual composition changes that might affect clinker quality are detected.
- A simple, easy-to-use web dashboard built with React.js or Angular for viewing results and downloading reports.

Premium Version

This advanced version targets large-scale operations with features like:

- Seamless integration with microscopy systems to enable real-time data transfer from INSEE’s optical microscopes.
- Advanced analytics for tracking trends in particle distribution and providing predictive insights to help optimize raw meal homogenization.
- Customizable alerts sent via SMS, email, or a dashboard to quickly notify staff about quality issues.
- API access for smooth integration with existing ERP or lab management systems, enabling automatic data exchange.
- Dedicated customer support for customized model training and fine-tuning based on the specific types of raw materials used.
- Scenario testing tools that allow users to analyse different raw meal compositions to find the best blending strategies.

Market Entry and Growth

- Pilot Implementation: Initially, the system will be deployed at INSEE Cement’s quality control lab to validate its performance with real raw meal samples and gather valuable user feedback.
- Partnerships: Collaborations will be pursued with other cement manufacturers and industry bodies in Sri Lanka to encourage wider adoption across the region.
- Pricing Model: The product will be offered via subscription, with tiered pricing that fits the scope of the community (basic) and premium (feature-rich) versions to accommodate different plant sizes and budgets.
- Continuous Improvement: Ongoing updates will be provided, driven by user feedback, and new machine learning models and image processing methods will be added to keep up with evolving industry requirements.

8. Reference list

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9. APPENDICIES

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